

4. SYSTEM FEATURES AND USER INTERACTION FLOWS

4.1 Administrative Staff Management

Authentication Process: Admin staff log in using institutional credentials. The system invokes `User.authenticate()` to verify credentials, performs role-based access control (RBAC) checks, and grants access to the administrative dashboard upon successful authentication.

Student Admission Workflow: Administrators capture comprehensive student information through a multi-section admission form including personal details, academic background, programme selection, and supporting documentation. Upon submission, the system creates two linked database records: a users table entry (with email, hashed password, and "Student" role) and a students table entry (with matriculation number, programme, level, and academic information). The system validates email uniqueness, required fields, and data formats before allowing submission.

Faculty Management: Admin staff add faculty members by creating user accounts with "Faculty" role and recording academic credentials, employment details, and department assignments. The system maintains comprehensive faculty profiles that can be updated for department transfers, promotions, or status changes.

Course Catalog Management: Administrators create courses by defining course codes, titles, credit hours, and prerequisites. They schedule course offerings for specific semesters, assign instructors (creating courseassignments records), set enrollment capacities, and define meeting schedules.

Programme Structure: Administrators define academic programmes and link courses through the programmerequirements table, designating courses as Core or Elective and assigning them to specific academic levels.

Student Record Management (CRUD): Administrators can search, view, edit, and deactivate student records. They can process programme transfers, level progressions, and status changes. The system uses soft deletion (setting Status to "Inactive") rather than permanent deletion to preserve historical records.

4.2 Student Self-Service Portal

Student Dashboard: After authentication, students access a personalized dashboard showing current semester information, academic standing (GPA, credits earned), important dates, and quick action links.

Course Registration System: When students initiate registration, the system executes a multi-step eligibility determination process: retrieves student profile (programme and level), queries programme requirements to identify applicable courses, cross-references with course offerings to show available courses, filters out previously completed courses, and evaluates prerequisite status. The system presents eligible courses with detailed information including instructor, schedule, enrollment status, and prerequisites.

Prerequisite Enforcement: The system queries the prerequisites table and verifies prerequisite completion in the student's academic history. Courses with unmet prerequisites are disabled with informative error messages. The system also enforces credit hour limits, preventing registration beyond the maximum allowable load.

Registration Confirmation: Students review selected courses before submission. The system performs final validations, creates student courses records (with Status "Registered"), updates enrollment counts, and sends confirmation emails.

Grade Viewing: Students access grades through a semester-by-semester view showing courses, grades, credit hours, and GPA calculations. The system displays cumulative statistics including overall GPA, total credits earned, and academic standing. Students can generate and print unofficial transcripts in PDF format, and request official transcripts through an administrative request system.

4.3 Faculty Teaching and Grading Tools

Faculty Dashboard: Faculty members view assigned courses (queried from courseassignments table), enrolled student counts, meeting schedules, and grade submission status. The dashboard displays visual indicators for courses requiring attention.

Grade Entry Workflow: Faculty select a course and view the class roster (students with Status "Registered" or "Completed"). They can enter grades through inline entry fields or dedicated grade entry modals. The system supports letter grades (A, B+, B, C+, C, D+, D, F) with automatic conversion to grade points.

Grade Processing: When faculty submit grades, the system invokes `assignGrade(studentId, courseId, grade)`, which validates inputs, checks for existing grade records, performs INSERT (new grade) or UPDATE (grade modification), updates studentcourses Status to "Completed", and records timestamps. All operations occur within a database transaction to maintain data consistency.

GPA Calculation: The system calculates GPA using the formula: $GPA = (\text{Sum of Quality Points}) / (\text{Sum of Credit Hours})$, where $\text{Quality Points} = \text{Grade Points} \times \text{Credit Hours}$. It calculates both term GPA (for a specific semester) and cumulative GPA (across all semesters), handling special cases like repeated courses and withdrawn courses.

Reporting Tools: Faculty can generate grade distribution reports, class performance summaries, and student performance alerts. Export functions support CSV/Excel and PDF formats for external record-keeping.